What Drives the Gain from Directional Noise?

Typical logical-sector gaps and rare ambiguous records in a charge-informed decoder

1. Scientific objective

Exact signed local charges carry more information than a parity syndrome. A preferred direction of pair creation can change that information even when the rate of active edges is fixed. Determine the mechanism by which direction bias changes optimal logical recovery in the bounded-current channel below.

The principal target is the physical distribution of logical-sector posterior gaps as the square patch grows. At $p = 0.30$, determine whether the observed changes in logical risk are explained by motion of typical gaps, by the probability and severity of rare ambiguous records, or by a finite-size crossover that the available evidence cannot resolve. Compare zero field, several finite fields, and the fully directed endpoint. Supply quantitative constraints and predictions that distinguish these possibilities; proving the thermodynamic phase is not required.

2. Geometry and logical recovery

For integer $L \geq 3$, the vertices are (i, j) with $0 \leq i, j \leq L - 1$. Retain all horizontal edges $((i, j), (i + 1, j))$, and retain vertical edges $((i, j), (i, j + 1))$ only for $1 \leq i \leq L - 2$. Edges are undirected for binary errors. The two rough boundaries are the columns $i = 0$ and $i = L - 1$. All other vertices are measured detectors. No observation is made at either rough boundary.

There are $2(L - 1)^2$ retained edges and $L(L - 2)$ detectors. Interior detectors have degree four; detectors on the top and bottom have degree three. No vertical boundary edge is an error variable. The shortest left-to-right logical path has $L - 1$ edges.

Let H be the detector-edge incidence matrix over $\mathbb{F}_2$. Let Γ_R contain exactly the retained edges meeting the right rough boundary. For any binary edge vector z, define

$$\ell(z) = \bigoplus_{e \in \Gamma_R} z_e.$$

A decoder returns a binary correction c on the retained edges. On binary activity vector x, it succeeds exactly when $Hc = Hx$ and $\ell(x \oplus c) = 0$. Exact reconstruction of x is unnecessary. Array ordering is your choice, but must be documented and exposed by the callable interface.

3. A bounded signed-current experiment

Orient horizontal edges toward increasing i and vertical edges toward increasing j. Independently on every retained edge, draw $j_e \in \{-1, 0, +1\}$ with

$$\Pr(j_e = 0) = 1 - p, \qquad \Pr(j_e = \pm 1) = \frac{pe^{\pm h}}{2\cosh h}.$$

The link field satisfies $0 \le h < \infty$; at $h = \infty$ use probabilities $(1 - p, p, 0)$ for $(0, +1, -1)$. At $h = 0$ the two nonzero currents are equally probable.

Define $x_e = \mathbf{1}\{j_e \neq 0\}$. The physical edge-activity rate is p at every field.

Let D be the integer incidence matrix on measured detectors, with -1 at the tail and $+1$ at the head of an edge. The complete public record is the exact signed interior charge vector

$$Q = Dj, \qquad s = Q \bmod 2 = Hx.$$

No charge is measured at a rough boundary. The decoder is told (G_L, p, h, Q) and receives no currents, activity bits, unmeasured boundary charges, or microscopic histories. The signs of Q are part of the record. There is only one observation round and no charge readout error.

The principal fields are $h = 0$, $\frac{1}{2}\log 9$, $\frac{1}{2}\log(47/3)$, $\frac{1}{2}\log(97/3)$, and $h = \infty$. Equivalently, the conditional probabilities of a positive current given activity are $1/2$, 0.90, 0.94, 0.97, and 1. These probabilities describe physical direction bias, not herald availability. You may add fields or physical rates where they improve the scientific conclusion. Include a lower-noise control at some $0 < p \le 0.05$.

4. Logical-sector posterior and the physical average

For a supported public record Q, define two absolute logical-sector weights

$$Z_\lambda(Q) = \sum_{j\,:\,Dj = Q,\ \ell(\mathbf{1}\{j \neq 0\}) = \lambda} \prod_e w_{p,h}(j_e), \qquad \lambda \in \{0,1\},$$

where $w_{p,h}$ is the edge law in Section 3. Set $q_\lambda(Q) = Z_\lambda(Q)/(Z_0(Q) + Z_1(Q))$. A logical Bayes decoder selects the more probable sector and returns a correction of that sector with syndrome s. A correction need not reconstruct the signed current.

Define

$$\Delta F(Q) = \log \frac{Z_0(Q)}{Z_1(Q)}, \qquad G(Q) = |\Delta F(Q)|.$$

If one sector has zero mass, set $G = \infty$ and its conditional Bayes risk to zero. Unsupported records do not enter a physical average. The optimal unconditional risk is

$$R_L^*(p,h) = \mathbb{E}_{Q \sim P_{L,p,h}}\left[\frac{1}{1 + e^{G(Q)}}\right].$$

The expectation is under the record distribution generated by the experiment. Conditional records, uniform record counting, selected low-charge records, and squared record probabilities define different averages.

5. Required scientific investigation

Construct and validate a method for logical-sector probabilities or controlled intervals for them. Establish independent exact checks on small patches. Investigate increasing genuinely two-dimensional patches, including sizes beyond direct whole-current enumeration. A narrow strip can be a validation tool but cannot replace the stated square-size comparison.

For the principal field comparison at $p = 0.30$, report unconditional Bayes-risk estimates, the distribution of G, and the contribution of ambiguous records to the risk. Include suitable gap quantiles and a quantitative lower-tail diagnostic. Explain how the selected diagnostics distinguish a changing typical record from a changing rare-record contribution. Numerical posterior error must be small enough, or bounded tightly enough, to support the comparison being made.

6. Mechanism, prediction, and limitations

State competing explanations before interpreting the final size trends. Compare their predictions using more than a single fitted risk curve. Reserve at least one size or field comparison for a check after the main analysis choices have been fixed, and report that check even if it contradicts the preferred explanation. You choose the sizes, sample counts, estimators, analytical bounds, and fitting or nonparametric methods.

Your result must identify what the data support about the risk mechanism and what they leave unresolved. A finite-size decline does not by itself establish exponential recovery or a sharp threshold. A count of configurations compatible with both sectors does not determine how their posterior masses balance. Neither an average gap nor an isolated ambiguous record is a substitute for the physical mean risk. A statistical-mechanics mapping must retain the channel, boundary convention, and binary logical target defined here.

Compare at least one practical charge-informed decoder with the controlled optimal-inference reference on the same public records. This comparison should determine how much of an observed practical trend is intrinsic to the channel and how much is algorithmic. Keep approximation error in the reference distinct from approximation in the practical decoder.

The task does not require a complete (p, h) phase diagram, a proof of a critical point, or a particular universality class. A conventional transition, slow crossover, or unresolved asymptotic behavior may each be a valid conclusion if supported by the stated experiment and uncertainty. Do not infer one channel's transition from a different distribution of unbounded currents or a different winding score.

7. Deliverables and resources

Submit a concise scientific report with the physical risk and gap-distribution results, practical-versus-intrinsic comparison, reserved prediction check, mechanism assessment, and limitations. Include sufficient data to recompute all physical averages, source, environment, and reproducible run instructions.

Provide `decode(graph, p, h, Q, rng=None) -> correction` and a callable sector-posterior estimator or certified-bound routine. Document orientation, incidence signs, graph and array ordering, numerical conventions for zero sector mass, and all randomness. Count missing or invalid corrections and inference failures explicitly; do not condition the report on successful convergence or silently omit hard records.

Design a study feasible on a scientific workstation. Justify the chosen sizes and precision against the available computational resources. Report actual time, memory, seeds, and stopping rules. Internet and standard scientific libraries are allowed. No particular decoder, gap scaling form, phase assignment, or critical exponent is prescribed.